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Labsheet-3

1. Write a Java program to input and display 5 integers using a 1-D array

```
import java.util.Scanner;

public class ArrayExample {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[] numbers = new int[5];

    // Input 5 integers
    System.out.println("Enter 5 integers:");
    for (int i = 0; i < 5; i++) {
      numbers[i] = sc.nextInt();
    }

    // Display the integers
    System.out.println("The entered integers are:");
    for (int i = 0; i < 5; i++) {
      System.out.println(numbers[i]);
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 5 integers:
10
20
30
40
50
```

The entered integers are:

10

20

30

40

50

2. Create a Java program to input 10 integers and display them in reverse order.

```
import java.util.Scanner;

public class ReverseArray {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[] numbers = new int[10];

    // Input 10 integers
    System.out.println("Enter 10 integers:");
    for (int i = 0; i < 10; i++) {
      numbers[i] = sc.nextInt();
    }

    // Display in reverse order
    System.out.println("Integers in reverse order:");
    for (int i = 9; i >= 0; i--) {
      System.out.println(numbers[i]);
    }

    sc.close();
  }
}
```

OUTPUT:-

Enter 10 integers:

1

2

3

4

5

6

7

8

9

10

Integers in reverse order:

10

9

8

7

6

5

4

3

2

1

3. Write a Java program to find the sum of all elements in a 1-D array.

```
import java.util.Scanner;

public class ArraySum {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
```

```
        int[] numbers = new int[5];
        int sum = 0;

        // Input array elements
        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
          numbers[i] = sc.nextInt();
        }

        // Calculate sum
        for (int i = 0; i < 5; i++) {
          sum = sum + numbers[i];
        }

        // Display sum
        System.out.println("Sum of all elements = " + sum);

        sc.close();
      }
    }
```

OUTPUT:-

```
Enter 5 integers:
10
20
30
40
50
Sum of all elements = 150
```

4. Create a Java program to find the maximum element in an array.

```
import java.util.Scanner;

public class MaximumElement {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[] numbers = new int[5];

    // Input array elements
```

```
        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        // Assume first element is maximum
        int max = numbers[0];

        // Find maximum element
        for (int i = 1; i < 5; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 5 integers:
12
45
23
67
34
Maximum element = 67
```

5. Write a Java program to count even and odd numbers in an array.

```
import java.util.Scanner;

public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];
        int even = 0;
        int odd = 0;
```

```
        // Input array elements
        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
          numbers[i] = sc.nextInt();
        }

        // Count even and odd numbers
        for (int i = 0; i < 5; i++) {
          if (numbers[i] % 2 == 0) {
            even++;
          } else {
            odd++;
          }
        }

        System.out.println("Number of even numbers = " + even);
        System.out.println("Number of odd numbers = " + odd);

        sc.close();
      }
    }
```

OUTPUT:-

```
Enter 5 integers:
10
15
22
31
40
Number of even numbers = 3
Number of odd numbers = 2
```

6. Create a Java program to copy all elements from one array to another.

```
import java.util.Scanner;
```

```
public class CopyArray {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[] arr1 = new int[5];
    int[] arr2 = new int[5];

    // Input elements into first array
    System.out.println("Enter 5 integers:");
    for (int i = 0; i < 5; i++) {
      arr1[i] = sc.nextInt();
    }

    // Copy elements from arr1 to arr2
    for (int i = 0; i < 5; i++) {
      arr2[i] = arr1[i];
    }

    // Display copied elements
    System.out.println("Elements of the second array:");
    for (int i = 0; i < 5; i++) {
      System.out.println(arr2[i]);
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 5 integers:
10
20
30
40
50

Elements of the second array:
10
20
```

30
40
50

7. Write a Java program to input and display elements of a 3 × 3 matrix.

```
import java.util.Scanner;

public class Matrix3x3 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[][] matrix = new int[3][3];

    // Input matrix elements
    System.out.println("Enter 9 elements of the matrix:");

    for (int i = 0; i < 3; i++) {
      for (int j = 0; j < 3; j++) {
        matrix[i][j] = sc.nextInt();
      }
    }

    // Display matrix
    System.out.println("The 3 x 3 matrix is:");

    for (int i = 0; i < 3; i++) {
      for (int j = 0; j < 3; j++) {
        System.out.print(matrix[i][j] + "\t");
      }
      System.out.println();
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
1
2
```

3
4
5
6
7
8
9

The 3 x 3 matrix is:
1 2 3
4 5 6
7 8 9
8. Create a Java program to find the sum of all elements in a matrix.

```
import java.util.Scanner;

public class MatrixSum {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];
      int sum = 0;

      // Input matrix elements
      System.out.println("Enter 9 elements of the matrix:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
            sum = sum + matrix[i][j];
         }
      }

      // Display matrix
      System.out.println("The matrix is:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            System.out.print(matrix[i][j] + "\t");
```

```
            }
            System.out.println();
        }

        // Display sum
        System.out.println("Sum of all elements = " + sum);

        sc.close();
    }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

The matrix is:

1 2 3

4 5 6

7 8 9

Sum of all elements = 45

9.Write a Java program to find the sum of each row in a matrix.

```
import java.util.Scanner;
```

```
public class RowSum {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];

      // Input matrix elements
      System.out.println("Enter 9 elements of the matrix:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
         }
      }

      // Find sum of each row
      for (int i = 0; i < 3; i++) {
         int sum = 0;

         for (int j = 0; j < 3; j++) {
            sum = sum + matrix[i][j];
         }

         System.out.println("Sum of row " + (i + 1) + " = " + sum);
      }
```

```
        sc.close();
    }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Sum of row 1 = 6

Sum of row 2 = 15

Sum of row 3 = 24

10. Create a Java program to find the sum of each column in a matrix.

```
import java.util.Scanner;

public class ColumnSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] matrix = new int[3][3];
```

```
        // Input matrix elements
        System.out.println("Enter 9 elements of the matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Find sum of each column
        for (int j = 0; j < 3; j++) {
            int sum = 0;

            for (int i = 0; i < 3; i++) {
                sum = sum + matrix[i][j];
            }

            System.out.println("Sum of column " + (j + 1) + " = " + sum);
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
1
```

2

3

4

5

6

7

8

9

Sum of column 1 = 12

Sum of column 2 = 15

Sum of column 3 = 18

11. Write a Java program to transpose a matrix.

```
import java.util.Scanner;

public class MatrixTranspose {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];

      // Input matrix elements
      System.out.println("Enter 9 elements of the matrix:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
```

```
        }
    }

    // Display original matrix
    System.out.println("Original Matrix:");

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            System.out.print(matrix[i][j] + "\t");
        }
        System.out.println();
    }

    // Display transpose
    System.out.println("Transpose of Matrix:");

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            System.out.print(matrix[j][i] + "\t");
        }
        System.out.println();
    }

    sc.close();
  }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Original Matrix:

1 2 3

4 5 6

7 8 9

Transpose of Matrix:

1 4 7

2 5 8

3 6 9

12. Create a Java program to add two matrices of the same size.

```
import java.util.Scanner;


public class MatrixAddition {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
```

```
int[][] matrix1 = new int[3][3];
int[][] matrix2 = new int[3][3];
int[][] sum = new int[3][3];

// Input first matrix
System.out.println("Enter elements of first matrix:");
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix1[i][j] = sc.nextInt();
   }
}

// Input second matrix
System.out.println("Enter elements of second matrix:");
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix2[i][j] = sc.nextInt();
   }
}

// Add two matrices
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      sum[i][j] = matrix1[i][j] + matrix2[i][j];
   }
}
```

```
        // Display result
        System.out.println("Sum of the two matrices:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(sum[i][j] + "\t");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter elements of first matrix:
1
2
3
4
5
6
7
8
9
```

Enter elements of second matrix:

9

8

7

6

5

4

3

2

1

Sum of the two matrices:

10 10 10

10 10 10

10 10 10

13. Write a Java program to multiply two matrices (basic case with 3 × 3 matrices).

```
import java.util.Scanner;

public class MatrixMultiplication {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix1 = new int[3][3];
      int[][] matrix2 = new int[3][3];
      int[][] result = new int[3][3];

      // Input first matrix
```

```
System.out.println("Enter elements of first matrix:");
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix1[i][j] = sc.nextInt();
   }
}

// Input second matrix
System.out.println("Enter elements of second matrix:");
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix2[i][j] = sc.nextInt();
   }
}

// Multiply the matrices
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      for (int k = 0; k < 3; k++) {
         result[i][j] += matrix1[i][k] * matrix2[k][j];
      }
   }
}

// Display result
System.out.println("Result of matrix multiplication:");
```

```
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(result[i][j] + "\t");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter elements of first matrix:
1
2
3
4
5
6
7
8
9
Enter elements of second matrix:
9
8
7
```

6

5

4

3

2

1

Result of matrix multiplication:

30 24 18

84 69 54

138 114 90

14. Write a Java program to input and display elements of a 2 × 2 × 2 3-D array.

```
import java.util.Scanner;

public class ThreeDArray {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][][] arr = new int[2][2][2];

      // Input elements
      System.out.println("Enter 8 elements:");

      for (int i = 0; i < 2; i++) {
         for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
               arr[i][j][k] = sc.nextInt();
```

```
            }
        }
    }

    // Display elements
    System.out.println("Elements of the 3-D array:");

    for (int i = 0; i < 2; i++) {
        System.out.println("Block " + (i + 1) + ":");

        for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
                System.out.print(arr[i][j][k] + "\t");
            }
            System.out.println();
        }
    }

    sc.close();
  }
}
```

OUTPUT:-

Enter 8 elements:

1

2

3

4

5

6

7

8

Elements of the 3-D array:

Block 1:

1 2

3 4

Block 2:

5 6

7 8

15. Create a Java program to find the sum of all elements in a 3-D array.

```
import java.util.Scanner;

public class ThreeDArraySum {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][][] arr = new int[2][2][2];
      int sum = 0;

      // Input elements
      System.out.println("Enter 8 elements:");

      for (int i = 0; i < 2; i++) {
```

```
        for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
                arr[i][j][k] = sc.nextInt();
                sum = sum + arr[i][j][k];
            }
        }
    }

    // Display sum
    System.out.println("Sum of all elements = " + sum);

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 8 elements:
1
2
3
4
5
6
7
8
Sum of all elements = 36
```

16. Write a Java program to count positive and negative numbers in a 3-D array.

```
import java.util.Scanner;

public class PositiveNegative3D {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][][] arr = new int[2][2][2];
      int positive = 0;
      int negative = 0;

      // Input elements
      System.out.println("Enter 8 elements:");

      for (int i = 0; i < 2; i++) {
         for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
               arr[i][j][k] = sc.nextInt();

               if (arr[i][j][k] > 0) {
                  positive++;
               } else if (arr[i][j][k] < 0) {
                  negative++;
               }
            }
         }
      }
```

```
        // Display results
        System.out.println("Number of positive numbers = " + positive);
        System.out.println("Number of negative numbers = " + negative);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 8 elements:
10
-5
20
-8
15
-2
7
-3
Number of positive numbers = 4
Number of negative numbers = 4
```

17. Create a Java program to find the maximum element in a 3-D array.

```
import java.util.Scanner;

public class Maximum3DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
int[][][] arr = new int[2][2][2];

// Input elements
System.out.println("Enter 8 elements:");

for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      for (int k = 0; k < 2; k++) {
         arr[i][j][k] = sc.nextInt();
      }
   }
}

// Assume first element is maximum
int max = arr[0][0][0];

// Find maximum element
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      for (int k = 0; k < 2; k++) {
         if (arr[i][j][k] > max) {
            max = arr[i][j][k];
         }
      }
   }
```

```
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 8 elements:
10
25
7
40
15
30
5
20
Maximum element = 40
```

18. Write a Java program to search for an element in a 3-D array and display its position.

```
import java.util.Scanner;

public class Search3DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] arr = new int[2][2][2];
```

```
// Input elements
System.out.println("Enter 8 elements:");

for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      for (int k = 0; k < 2; k++) {
         arr[i][j][k] = sc.nextInt();
      }
   }
}

// Input element to search
System.out.print("Enter element to search: ");
int search = sc.nextInt();

boolean found = false;

// Search element
for (int i = 0; i < 2; i++) {
   for (int j = 0; j < 2; j++) {
      for (int k = 0; k < 2; k++) {
         if (arr[i][j][k] == search) {
            System.out.println("Element found at position: ["
                  + i + "][" + j + "][" + k + "]");
            found = true;
```

```
            }
          }
        }
      }

      if (!found) {
         System.out.println("Element not found.");
      }

      sc.close();
   }
}
```

OUTPUT:-

```
Enter 8 elements:
10
20
30
40
50
60
70
80
Enter element to search: 60
Element found at position: [1][0][1]
```

19. Write a Java program to create and display a jagged array with 3 rows of different lengths.

```
import java.util.Scanner;

public class JaggedArray {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      // Create jagged array with 3 rows
      int[][] arr = new int[3][];

      // Different lengths for each row
      arr[0] = new int[2];
      arr[1] = new int[3];
      arr[2] = new int[4];

      // Input elements
      System.out.println("Enter elements:");

      for (int i = 0; i < arr.length; i++) {
         System.out.println("Enter " + arr[i].length + " elements for row " + (i + 1) + ":");

         for (int j = 0; j < arr[i].length; j++) {
            arr[i][j] = sc.nextInt();
         }
      }

      // Display jagged array
```

```
        System.out.println("Jagged Array:");

        for (int i = 0; i < arr.length; i++) {
            for (int j = 0; j < arr[i].length; j++) {
                System.out.print(arr[i][j] + "\t");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter elements:
Enter 2 elements for row 1:
10
20
Enter 3 elements for row 2:
30
40
50
Enter 4 elements for row 3:
60
70
80
90
```

Jagged Array:

10 20

30 40 50

60 70 80 90

20. Create a Java program to store marks for students where each student has taken a different number of subjects, then display them.

```
import java.util.Scanner;

public class StudentMarks {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      // Jagged array: different subjects for each student
      int[][] marks = new int[3][];

      // Different number of subjects
      marks[0] = new int[3];
      marks[1] = new int[4];
      marks[2] = new int[2];

      // Input marks
      for (int i = 0; i < marks.length; i++) {
         System.out.println("Enter marks for Student " + (i + 1) + ":");

         for (int j = 0; j < marks[i].length; j++) {
            System.out.print("Subject " + (j + 1) + ": ");
```

```
            marks[i][j] = sc.nextInt();
        }
    }

    // Display marks
    System.out.println("\nStudent Marks:");

    for (int i = 0; i < marks.length; i++) {
        System.out.print("Student " + (i + 1) + ": ");

        for (int j = 0; j < marks[i].length; j++) {
            System.out.print(marks[i][j] + " ");
        }

        System.out.println();
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter marks for Student 1:
Subject 1: 80
Subject 2: 75
Subject 3: 90
```

Enter marks for Student 2:

Subject 1: 85

Subject 2: 70

Subject 3: 88

Subject 4: 92

Enter marks for Student 3:

Subject 1: 78

Subject 2: 84

Student Marks:

Student 1: 80 75 90

Student 2: 85 70 88 92

Student 3: 78 84

21. Write a program to rotate an array to the left by one position.

```
import java.util.Scanner;

public class LeftRotate {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[] arr = new int[5];

      // Input array elements
      System.out.println("Enter 5 integers:");
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }
```

```
        // Store the first element
        int first = arr[0];

        // Shift elements to the left
        for (int i = 0; i < 4; i++) {
            arr[i] = arr[i + 1];
        }

        // Put first element at the end
        arr[4] = first;

        // Display rotated array
        System.out.println("Array after left rotation:");
        for (int i = 0; i < 5; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 5 integers:
10
20
30
```

40

50

Array after left rotation:

20 30 40 50 10

22. Count how many times each element appears in a 1-D array.

```
import java.util.Scanner;

public class ElementFrequency {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[] arr = new int[5];
      boolean[] visited = new boolean[5];

      // Input array elements
      System.out.println("Enter 5 integers:");
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }

      // Count frequency of each element
      System.out.println("Frequency of each element:");

      for (int i = 0; i < 5; i++) {

         if (visited[i]) {
```

```
            continue;
        }

        int count = 1;

        for (int j = i + 1; j < 5; j++) {
            if (arr[i] == arr[j]) {
                count++;
                visited[j] = true;
            }
        }

        System.out.println(arr[i] + " appears " + count + " time(s)");
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 5 integers:
10
20
10
30
20
```

Frequency of each element:

10 appears 2 time(s)

20 appears 2 time(s)

30 appears 1 time(s)

23. Find the second largest element in a 1-D array.

```
import java.util.Scanner;

public class SecondLargest {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[] arr = new int[5];

      // Input array elements
      System.out.println("Enter 5 integers:");
      for (int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }

      int largest = Integer.MIN_VALUE;
      int secondLargest = Integer.MIN_VALUE;

      // Find largest and second largest
      for (int i = 0; i < 5; i++) {
         if (arr[i] > largest) {
            secondLargest = largest;
```

```
            largest = arr[i];
        } else if (arr[i] > secondLargest && arr[i] != largest) {
            secondLargest = arr[i];
        }
    }

    if (secondLargest == Integer.MIN_VALUE) {
        System.out.println("Second largest element does not exist.");
    } else {
        System.out.println("Second largest element = " + secondLargest);
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 5 integers:
10
45
25
60
30
Second largest element = 45
```

24. Remove duplicate elements from a 1-D array (without using collections).

```
import java.util.Scanner;
```

```
public class RemoveDuplicates {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];
        int[] unique = new int[5];
        int count = 0;

        // Input array elements
        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        // Remove duplicate elements
        for (int i = 0; i < 5; i++) {
            boolean duplicate = false;

            for (int j = 0; j < count; j++) {
                if (arr[i] == unique[j]) {
                    duplicate = true;
                    break;
                }
            }

            if (!duplicate) {
```

```
            unique[count] = arr[i];
            count++;
        }
    }

    // Display array without duplicates
    System.out.println("Array after removing duplicates:");

    for (int i = 0; i < count; i++) {
        System.out.print(unique[i] + " ");
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter 5 integers:
10
20
10
30
20
Array after removing duplicates:
10 20 30
```

25. Find the sum of the main diagonal and secondary diagonal elements of a square matrix.

```
import java.util.Scanner;
```

```
public class DiagonalSum {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];
      int mainSum = 0;
      int secondarySum = 0;

      // Input matrix elements
      System.out.println("Enter 9 elements of the matrix:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
         }
      }

      // Calculate diagonal sums
      for (int i = 0; i < 3; i++) {
         mainSum = mainSum + matrix[i][i];
         secondarySum = secondarySum + matrix[i][2 - i];
      }

      // Display matrix
      System.out.println("Matrix:");
```

```
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + "\t");
            }
            System.out.println();
        }

        System.out.println("Sum of main diagonal = " + mainSum);
        System.out.println("Sum of secondary diagonal = " + secondarySum);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
1
2
3
4
5
6
7
8
9
```

Matrix:

1 2 3

4 5 6

7 8 9

Sum of main diagonal = 15

Sum of secondary diagonal = 15

26. Display the upper and lower triangular parts of a square matrix separately.

```
import java.util.Scanner;

public class TriangularMatrix {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];

      // Input matrix elements
      System.out.println("Enter 9 elements of the matrix:");

      for (int i = 0; i < 3; i++) {
         for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
         }
      }

      // Display Upper Triangular Matrix
      System.out.println("Upper Triangular Matrix:");
```

```
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        if (i <= j) {
            System.out.print(matrix[i][j] + "\t");
        } else {
            System.out.print("0\t");
        }
    }
    System.out.println();
}

// Display Lower Triangular Matrix
System.out.println("Lower Triangular Matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        if (i >= j) {
            System.out.print(matrix[i][j] + "\t");
        } else {
            System.out.print("0\t");
        }
    }
    System.out.println();
}
```

```
        sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
1
2
3
4
5
6
7
8
9
Upper Triangular Matrix:
1       2       3
0       5       6
0       0       9
Lower Triangular Matrix:
1       0       0
4       5       0
7       8       9
```

27. Print only the boundary elements of a matrix.

```
import java.util.Scanner;

public class BoundaryElements {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int[][] matrix = new int[3][3];

    // Input matrix elements
    System.out.println("Enter 9 elements of the matrix:");

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            matrix[i][j] = sc.nextInt();
        }
    }

    // Display only boundary elements
    System.out.println("Boundary elements:");

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (i == 0 || i == 2 || j == 0 || j == 2) {
                System.out.print(matrix[i][j] + "\t");
            } else {
                System.out.print(" \t");
            }
        }
        System.out.println();
```

```
        }


        sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
1
2
3
4
5
6
7
8
9
Boundary elements:
1       2       3
4               6
7       8       9
```

28. Search for an element in a 2-D array and display its row & column position.

```
 import java.util.Scanner;


public class Search2DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
int[][] matrix = new int[3][3];

// Input matrix elements
System.out.println("Enter 9 elements of the matrix:");

for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix[i][j] = sc.nextInt();
   }
}

// Input element to search
System.out.print("Enter element to search: ");
int search = sc.nextInt();

boolean found = false;

// Search the element
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      if (matrix[i][j] == search) {
         System.out.println("Element found at row "
               + (i + 1) + ", column " + (j + 1));
         found = true;
      }
```

```
        }
      }

      if (!found) {
        System.out.println("Element not found.");
      }

      sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
10
20
30
40
50
60
70
80
90
Enter element to search: 60
Element found at row 2, column 3
```

29. Sort the elements of each row in a jagged array.

```
import java.util.Scanner;
```

```
public class SortJaggedArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Create jagged array with 3 rows
        int[][] arr = new int[3][];

        // Different lengths for each row
        arr[0] = new int[3];
        arr[1] = new int[4];
        arr[2] = new int[2];

        // Input elements
        System.out.println("Enter elements:");

        for (int i = 0; i < arr.length; i++) {
            System.out.println("Enter " + arr[i].length +
                    " elements for row " + (i + 1) + ":");

            for (int j = 0; j < arr[i].length; j++) {
                arr[i][j] = sc.nextInt();
            }
        }

        // Sort each row using bubble sort
        for (int i = 0; i < arr.length; i++) {
```

```
        for (int j = 0; j < arr[i].length - 1; j++) {
            for (int k = 0; k < arr[i].length - j - 1; k++) {
                if (arr[i][k] > arr[i][k + 1]) {
                    int temp = arr[i][k];
                    arr[i][k] = arr[i][k + 1];
                    arr[i][k + 1] = temp;
                }
            }
        }
    }

    // Display sorted jagged array
    System.out.println("\nJagged Array after sorting each row:");

    for (int i = 0; i < arr.length; i++) {
        for (int j = 0; j < arr[i].length; j++) {
            System.out.print(arr[i][j] + "\t");
        }
        System.out.println();
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter elements:
```

Enter 3 elements for row 1:

30

10

20

Enter 4 elements for row 2:

80

40

60

50

Enter 2 elements for row 3:

90

70

Jagged Array after sorting each row:

10 20 30

40 50 60 80

70 90

30. Check if a given matrix is a sparse matrix (more zero elements than non-zero).

```
import java.util.Scanner;

public class SparseMatrix {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] matrix = new int[3][3];
      int zeroCount = 0;
      int nonZeroCount = 0;
```

```
// Input matrix elements
System.out.println("Enter 9 elements of the matrix:");

for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      matrix[i][j] = sc.nextInt();

      if (matrix[i][j] == 0) {
         zeroCount++;
      } else {
         nonZeroCount++;
      }
   }
}

// Display matrix
System.out.println("Matrix:");

for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      System.out.print(matrix[i][j] + "\t");
   }
   System.out.println();
}
```

```
        // Check sparse matrix
        if (zeroCount > nonZeroCount) {
            System.out.println("The matrix is a Sparse Matrix.");
        } else {
            System.out.println("The matrix is not a Sparse Matrix.");
        }

        System.out.println("Number of zero elements = " + zeroCount);
        System.out.println("Number of non-zero elements = " + nonZeroCount);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter 9 elements of the matrix:
0
0
5
0
2
0
0
0
3
Matrix:
```

0 0 5

0 2 0

0 0 3

The matrix is a Sparse Matrix.

Number of zero elements = 6

Number of non-zero elements = 3